

Homework 3: Iteration

1.

a.

```
int i = 0;
while ( i < n ){
    int result = i * i;
    out.print ( result + " ");
    i++;
}
```

b.

```
int i = 10;
while ( i < n ){
    out.print ( i + " ");
    i = i + 10;
}
```

c.

```
int i = 1;
while ( i < n ){
    out.print ( i + " ");
    i = i * 2;
}
```

2.

a.

Statement	Variable Values
	<pre>i = 0 j = 10 n = 0</pre>
while (i < j) {	
	<pre>i = 0 1 2 3 4 j = 10 9 8 7 6 n = 0 1 2 3 4</pre>
i++;	
	<pre>i = 1 2 3 4 5 j = 10 9 8 7 6 n = 0 1 2 3 4</pre>
j--;	
	<pre>i = 1 2 3 4 5 j = 9 8 7 6 5 n = 0 1 2 3 4</pre>
n++;	
	<pre>i = 1 2 3 4 5 j = 9 8 7 6 5 n = 1 2 3 4 5</pre>

}	
	i = 5 j = 5 n = 5

b.

Statement	Variable Values
	i = 0 j = 0 n = 0
while (i < 10) {	
	i = 0 1 2 3 4 5 6 7 8 9 j = 0 1 2 3 4 5 6 7 8 9 n = 0 1 4 9 16 25 36 49 64 81
i++;	
	i = 1 2 3 4 5 6 7 8 9 10 j = 0 1 2 3 4 5 6 7 8 9 n = 0 1 4 9 16 25 36 49 64 81
n = n + i + j;	
	i = 1 2 3 4 5 6 7 8 9 10 j = 0 1 2 3 4 5 6 7 8 9 n = 1 4 9 16 25 36 49 64 81 100
j++;	
	i = 1 2 3 4 5 6 7 8 9 10 j = 1 2 3 4 5 6 7 8 9 10 n = 1 4 9 16 25 36 49 64 81 100
}	
	i = 10 j = 10 n = 10

c.

Statement	Variable Values
	i = 10 j = 0 n = 0
while (i > 0) {	
	i = 10 9 8 7 6 5 4 3 2 1 j = 0 1 2 3 4 5 6 7 8 9 n = 0 8 14 18 20 20 18 14 8 0
i--;	
	i = 9 8 7 6 5 4 3 2 1 0 j = 0 1 2 3 4 5 6 7 8 9

	n = 0 8 14 18 20 20 18 14 8 0
j++;	
	i = 9 8 7 6 5 4 3 2 1 0 j = 1 2 3 4 5 6 7 8 9 10 n = 0 8 14 18 20 20 18 14 8 0
n = n + i - j;	
	i = 9 8 7 6 5 4 3 2 1 0 j = 1 2 3 4 5 6 7 8 9 10 n = 8 14 18 20 20 18 14 8 0 -10
}	
	i = 0 j = 10 n = -10

d.

Statement	Variable Values
	i = 0 j = 10 n = 0
while (i != j) {	
	i = 0 2 4 j = 10 8 6 n = 0 1 2
i = i + 2;	
	i = 2 4 6 j = 10 8 6 n = 0 1 2
j = j - 2;	
	i = 2 4 6 j = 8 6 4 n = 0 1 2
n++;	
	i = 2 4 6 j = 8 6 4 n = 1 2 3
}	
	i = ? j = ? n = ? i will never equal to j.

e.

Statement	Variable Values
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	i = 3 j = 4 n = 0
while (i != 0) {	
	i = 3 2 1 j = 4 4 4 n = 0 4 8
n += j;	
	i = 3 2 1 j = 4 4 4 n = 4 8 12
i--;	
	i = 2 1 0 j = 4 4 4 n = 4 8 12
}	
	i = 0 j = 4 n = 12

3.

```
int s = 0;
int i = 1;
while ( i <= 10 ){
    s = s + 1;
    i++;
}
```
4.

```
int i = 0;
double term = 0;
while ( i < n ) {
    term = Math.pow ( -1, i ) / ( 2 * i + 1 );
    pi = pi + term;
    i++;
}
pi = pi * 4;
```
5.

```
int n = 1;
while ( n * n < areabound ) {
    int m = 1;
    while ( m * m < areabound ){
        sum = sum + n * n + m * m;
        m++;
    }
    n++;
}
```

```
}
```

Additional Question:

```
1.  int i = 0;
    double term = 0;
    double pi = 0;
    double previous = 100;
    double error = 0.0001;
    while (Math.abs(pi - previous) >= error) {
        previous = pi;
        term = Math.pow( -1, i ) / ( 2 * i + 1);
        pi = pi + term;
        i++;
    }
    pi = pi * 4;
```